

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer-2023**

Course: B. Tech. Branch : Electronics & Telecommunication Engineering
Subject Code & Name: BTETPE504A Analog Circuits Semester : V
Max Marks: 60 Date:17/08/2023 Duration: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) How does a non-inverting amplifier configuration work with operational amplifiers? Provide an analysis of its voltage gain and input/output characteristics.	Level 1	6
B) What is the significance of the Bode plot in operational amplifier circuits? How does it help to understand the frequency response?	Level 1	6
C) Explain the concept of negative feedback in operational amplifiers. How does it affect the performance of the amplifier?	Level 2	6
Q.2 Solve Any Two of the following.		12
A) Explain the working principle of an inverting amplifier using op-amp.	Level 3	6
B) Explain how op-amp can be used as an integrator?	Level 2	6
C) Design a summing amplifier using an operational amplifier that can add multiple input signals with different gains.	Level 1	6
Q. 3 Solve Any Two of the following.		12
A) Explain the working principle of a comparator using an operational amplifier. How does it compare two input signals and produce a digital output?	Level 1	6
B) What are clippers and clampers, and how are they implemented using operational amplifiers?	Level 1	6
C) What is a logarithmic amplifier, and how is it implemented with operational amplifiers? Describe its application in signal processing and measurement.	Level 2	6
Q.4 Solve Any Two of the following.		12
A) Explain a Schmitt trigger oscillator using operational amplifiers.	Level 1	6
B) Explain a Relaxation oscillator using operational amplifiers.	Level 1	6
C) Explain the importance of frequency synchronization and phase locking in certain applications of waveform generation using multiple oscillators.	Level 1	6
Q. 5 Solve Any Two of the following.		12
A) Compare RC oscillators with LC oscillators.	Level 1	6
B) Explain the advantages and disadvantages of flash-type ADCs.	Level 2	6
C) Explain the operation of a voltage-to-current (V-I) converter.	Level 1	6

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